Layered representation diagnostics for short metagenomic reads

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Abstract

Short-read metagenomic pipelines compress trimmed or perturbed reads before classification or database lookup, but the retained sequence attributes are rarely audited directly. We present a representation-diagnostic framework spanning every integer length from 50 to 75 bp, with 100, 125 and 150 bp anchors. CK4P-MSP is a 222-dimensional, training-free representation combining reverse-complement canonical 4-mer composition (K), global nucleotide-property-coded summaries (P) and multi-scale positional property pooling (MSP). A seven-group ablation associated P with global stability, MSP with grouped local-change readout and K with composition-linked retrieval. Across the controlled whole-genome-sequence perturbation grid, CK4P-MSP achieved mean paired cosine 0.989, standardized drift 0.129 and 0.979 macro-F1 in the primary template-grouped local-versus-nuisance readout. The 50–75 bp shared-template sweep preserved the same stability–readout ordering. In six source-grouped CAMI target-versus-background tasks, a probe trained on 100-bp clean reads gave shifted macro-F1 0.793, comparable to CK4/CK5 and descriptively 0.010–0.012 above PseKNC/PseEIIP. In a separate historical-descriptor local-readout audit, PseKNC had lower drift whereas CK4P-MSP retained stronger structured local-change readout. Train-fitted coordinate scaling exposed an N-mask transfer failure. Full-position encodings provided higher-dimensional positional-readout references, and compressed high-k controls were less stable at the same feature budget. CK4P-MSP therefore defines a block-attributable trade-off for perturbation audits alongside database-linked matching.

Keywords metagenomics, short reads, representation diagnostics, k-mer features, nucleotide-property coding

Metagenomic next-generation sequencing can interrogate complex or unexpected microbial mixtures without a fixed target panel, and it has become an important route for infectious-disease investigation when targeted assays are insufficient [1, 2, 3]. Yet the reads entering downstream analysis are often short, trimmed, ambiguous and locally corrupted. Adapter removal, quality trimming and low-biomass contamination can all change the effective sequence evidence available to a pipeline before classification begins [4, 5, 6]. Read length can affect homolog recovery and taxonomic assignment, although the magnitude and direction of the effect depend on the classifier, database, taxonomic group and error profile [7, 8, 9, 10]. Reduced context can narrow the redundancy of local sequence evidence and make it harder to determine which information categories remain available. A read representation may already have compressed local composition, biochemical regularity or positional information before a downstream benchmark reports success or failure. Short-read representation therefore defines which categories of evidence remain available to a later classifier, alignment routine or database query.

Many metagenomic workflows rely on exact or near-exact match evidence. Canonical k-mer, minimizer, alignment and database-backed systems such as *Kraken*-family tools, *Centrifuge*, *CLARK*, *Kaiju* and related indexing frameworks have established the practical value of explicit sequence matching for taxonomic assignment and downstream biological interpretation [9, 11, 12, 13, 10]. Community benchmarks further show that performance depends on database coverage, sample composition, novelty and evaluation context, not only on the classifier name [14, 15]. Database-linked match evidence remains the appropriate backbone for species discrimination, allele-sensitive interpretation and resistance-gene calling. Spaced-seed indexing extends this line by tuning match patterns for sensitivity under mutation [16]. Alignment-free comparison adds a related interpretive problem: similarity scores are affected by sequence length and require objective functions or score-distribution analyses before they can be assigned biological meaning [17]. Additional audit coordinates may therefore be especially informative toward the lower end of the 50–75 bp range, where local matching evidence has less redundancy than at 150 bp. These systems identify database matches, whereas the present study characterizes how composition, nucleotide-property-coded and positional signals change within compact representations when reads are short, noisy or partially degraded.

Vectorized k-mer representations provide the first broad alternative to explicit matching. Here the central operation is to collapse each read or fragment into a composition profile, such as count, frequency, relative-frequency, TF-IDF, or background-adjusted features. PlasFlow and the BMC Bioinformatics 2018 bacteria classifier show that this compact route can be highly competitive when the task is composition-driven rather than order-driven [18, 19]. Resource-saving k-mer classification extends the same logic by showing that low-dimensional k-mer distributions plus classical machine learning can remain competitive [20]. Recent cross-genome analysis of k-mer rarity further shows that k-mer length changes the balance between abundance, compositional bias and taxonomic specificity [21]. A complementary study used both bulk and positional k-mers in an interpretable mixture model, demonstrating that composition and positional organization can support different sequence analyses [22]. These studies motivate the present distinction between a compact composition backbone and a separate coarse positional summary. They do not, however, establish how nucleotide-property-coded positional channels behave under matched short-read perturbations. Accordingly, vectorized k-mer methods are treated here as composition-focused comparators whose performance depends on how much local composition survives compression in the short-read regime.

Hashing and sketching form a separate compression family rather than a lower-rank version of singular-value decomposition (SVD). MinHash, locality-sensitive hashing (LSH), histosketch and related strategies map k-mer sets or profiles into fixed-length signatures for rapid similarity search, large-scale database indexing or ultra-lightweight compression [23, 24, 25, 26]. Kssd and the HULK/histosketch line compress representations in service of indexing or neighbourhood search rather than preserving linear variance structure. The present analysis instead evaluates signal retention in compact vectors under fixed probes and perturbations. Sketching is therefore treated as a neighbouring compression paradigm and assessed in its native similarity geometry.

Biochemical and signal-based encodings add a second alternative representation family. Chaos-game and genomic-signal formulations, together with numerical mappings such as Voss, tetrahedron and electron-ion interaction potential (EIIP), expose sequence structure through algebraic or physicochemical coordinates rather than literal token identity alone [27, 28, 29, 30, 31, 32, 33]. Nucleotide chemical-property encodings have also represented ring structure, chemical functionality and hydrogen-bond class, often together with cumulative nucleotide density or other position-dependent frequencies [34]. These mappings have been used well beyond metagenomic classification, including promoter, regulatory-site, nucleosome-positioning and nucleotide-modification prediction. Their value is therefore not specific to mNGS or to k-mer classifiers. At the same time, the biological interpretation of a named property does not by itself establish that its numerical coordinate is causal, non-redundant or robust across sequence lengths and tasks.

Hybrid handcrafted descriptors already combine composition, physicochemical properties and sequence-order information. Pseudo k-tuple nucleotide composition (PseKNC) augments k-tuple frequencies with lagged correlations between oligonucleotide properties, and PseKNC-General and repDNA expose broad families of autocorrelation, pseudo-composition and user-defined physicochemical descriptors [35, 36, 37]. Pseudo electron-ion interaction potential (PseEIIP) similarly weights trinucleotide composition by EIIP, whereas platforms such as iFeatureOmega catalogue composition, position-specific, autocorrelation and physicochemical encoders within one feature-engineering system [38, 39]. Recent multiscale density encodings further show that local and broader position-dependent nucleotide patterns can be combined with learned models [40]. Learning-based multimodal models provide a related but distinct precedent: MIXBend aligns k-mer sequence features with nucleotide physicochemical and positional information in a self-supervised framework for 50-bp DNA bendability prediction [41]. These studies establish that composition, physicochemical and positional signals can be fused for task-specific prediction. The methodological gap is narrower: such representations are usually assessed through the performance of a task-specific predictor, whereas a common matched-perturbation audit that attributes stability and failure modes to individual blocks remains less developed. Here the contribution is a compact, training-free and block-decomposable representation-and-audit protocol for controlled short-read perturbations, rather than a claim that the underlying nucleotide properties or their fusion are new.

Sequence-token deep learning classifiers make the neighbouring problem space especially relevant. DeepMicrobes introduced deep learning for metagenomic taxonomic classification, using canonical k-mer tokenization, embedding, BiLSTM and attention for read-level assignment [42]. MetaTransformer extended this family with self-attention and alternative tokenization or embedding schemes, including character-level, BPE, k-mer and hash/LSH variants [43]. DeepMicroClass provides a related example of a task-specific deep classifier evaluated across sequence classes and external CAMI II data, although its primary unit is a metagenomic contig rather than a 50–150 bp read [44]. BERTax showed that a small-k Transformer can support hierarchical taxonomy when pretraining and output heads are aligned to the task [45]. BarcodeBERT further demonstrated that barcode-scale biodiversity tasks benefit from domain-specific pretraining and masking choices, although its setting is barcode classification rather than shotgun short-read mNGS [46]. Large pre-trained DNA language models extend this representation family as general-purpose encoders [47, 48, 49, 50, 51]. Such learned embeddings can provide strong predictive features, particularly when pretraining data and compute are available. These models provide a predictive-performance reference point, whereas the present study examines a low-dimensional, training-free and block-decomposable representation under controlled short-read perturbation.

Raw-sequence and position-aware models complete the picture. DeepVirFinder, Virtifier, and DETIRE demonstrate that CNN or hybrid architectures can extract useful motif- and composition-level signal from viral fragments, but their target is viral detection rather than multi-class short-read taxonomy [52, 53, 54]. KmerAperture adds an important boundary case by showing that ordinary k-mer set comparisons can lose synteny, whereas matching k-mers back to their original lists can retain positional information about core and accessory differences [55].

Collectively, the reviewed literature has established composition vectors, physicochemical descriptors, positional k-mers, task-specific hybrid encoders and learned sequence embeddings. These approaches answer different predictive or retrieval questions. A common protocol for measuring how compact composition, nucleotide-property-coded and coarse positional blocks retain, lose or redistribute signal under matched perturbations remains underdeveloped. The present work is therefore positioned as a representation-and-audit contribution that makes those trade-offs explicit, rather than as a replacement for the strongest task-specific predictor in any one application.

Against this background, we treat short-read representation as an audit problem with several distinct endpoints. The unmet methodological need is a fixed-budget protocol that attributes perturbation responses to local composition, nucleotide-property summaries and coarse relative position. We ask how local k-mer-composition retention, nucleotide-property-coded stability, positional readout accessibility and compactness trade off across controlled short-read settings. CK4P-MSP combines reverse-complement canonical 4-mer composition with global nucleotide-property-coded summaries and multi-scale property pooling to produce a block-decomposable audit coordinate system. The resulting working representation is a compact, training-free coordinate system for controlled short-read perturbation audits. Its methodological contribution is a block-level attribution protocol that measures the channels separately under matched perturbations, extending rather than originating composition–property fusion. Full-position matrices provide high-dimensional positional-readout references, whereas the canonical spaced-property (CSP) control tests the transfer of matching-oriented spaced-seed priors to dense read-level representations. We combine complete K/P/MSP ablation with matched perturbation, comparator and external-probe audits. A shared-template sweep samples every integer length from 50 to 75 bp, and the 100, 125 and 150 bp anchors assess whether lower-range trends persist as sequence context increases. The analysis estimates metric-specific conditional contributions and identifies the operating range of each block under a compact, block-attributable feature budget.

Materials and Methods

Study design

We designed the study as a representation-diagnostic analysis of controlled short-read perturbation settings. We define the primary audit profile through four outputs: paired stability, nearest-clean retrieval, grouped local-change readout and blockwise contribution profiles. Feature dimension is reported separately as a resource axis for compactness. The central question was whether nucleotide-property-coded and coarse positional summaries add auditable evidence to a canonical local k-mer-composition backbone. CK4 and CK5 were reverse-complement canonical k-mer-composition baselines; P denoted global nucleotide-property-coded summaries; MSP denoted multi-scale positional property pooling. The seven non-empty K/P/MSP combinations were evaluated, full-position matrices were treated as high-dimensional positional-readout references, and the canonical spaced-property (CSP) control served as a spaced-seed mechanism comparator.

This design makes two questions testable. First, whether a compact mixed representation remains close to its clean counterpart under nuisance perturbation while retaining representation-level readout accessibility. Second, whether the observed behaviour can be assigned to metric-specific K, P or MSP contributions rather than to dimensionality, normalization or dilution in a mixed L2 space. We therefore combined a seven-group ablation with a 2×2 spatial-localization-by-substitution-chemistry audit, representative historical handcrafted descriptors, same-dimension principal-component analysis (PCA) and singular-value decomposition (SVD) controls, dimension-matched high-k compressed k-mer baselines, block-weight and property-scaling sensitivity, paired non-parametric tests, k-nearest-neighbour estimator-sensitivity checks for empirical separability, P/MSP relation analysis, feature-extraction runtime and error-aware ART simulator strata. Shallow probes quantified the accessibility of retained signal to a fixed low-capacity model. Mixed-space L2 quantifies standardized representation drift in the declared block-normalized coordinates and has no direct interpretation in biophysical units.

For a new representation, the audit protocol reports these four outputs. Paired stability and composition-linked retrieval are computed on matched clean–perturbed pairs; grouped local-change readout uses template-grouped probes; and blockwise contribution profiles compare the corresponding K, P and MSP ablations and relation controls. Feature dimension accompanies the profile as a resource axis. Together, these outputs define an endpoint-specific profile of retained signal and observed failure modes.

The lower-range design was anchored to a commonly reported 50–75 bp mNGS read-length regime [56, p. 99]. A shared-template continuity audit evaluated every integer length in this interval by prefix-truncating the same 420 independently sampled 150-bp source windows. This design changed read length without changing source identity. Analyses requiring smaller condition sets retained prespecified subsets of these lower-range lengths for consistency across experimental layers. Broader anchors at 100, 125 and 150 bp tested whether lower-range behaviour persisted as more sequence context became available, and 300 bp was retained as an extended-read reference in selected boundary analyses. The design therefore samples 50–75 bp densely and uses 100, 125 and 150 bp as broader context anchors; individual lengths are analytical conditions rather than biological thresholds.

Representation families

The representation families were defined to separate local k-mer composition, global nucleotide-property-coded summaries and coarse positional information. Canonical 4-mer (CK4) and canonical 5-mer (CK5) blocks count reverse-complement canonical contiguous k-mers. P contains the read-level mean and population standard deviation of hydrogen-bond, GC, purine and electron-ion interaction potential (EIIP) maps, together with N fraction, length divided by 200 and Shannon entropy over A/C/G/T/N divided by $\log_2 5$. MSP mean-pools five per-base nucleotide-property maps (hydrogen bond, GC, purine, EIIP and N indicator) over relative-position binsets with 2, 3, 4 and 6 bins. For a read of length L and a scale with b bins, the jth bin spans indices $[\lfloor jL/b \rfloor, \lfloor (j+1)L/b \rfloor)$; any

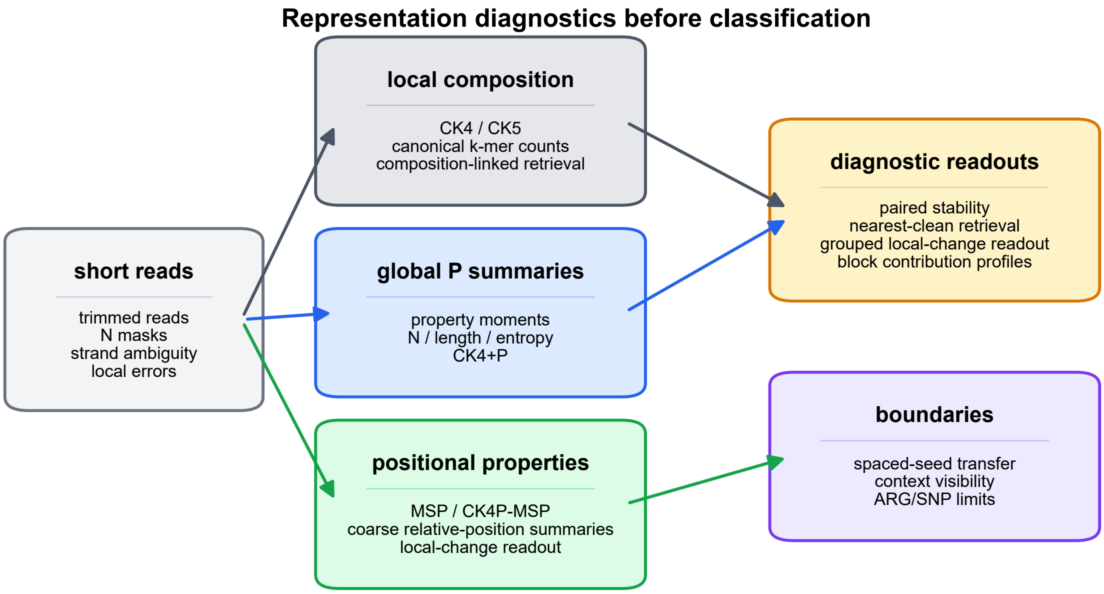

Figure 1 Representation-diagnostic framework and read-length regime. Clean and perturbed reads are encoded through local k-mer composition (K), global property summaries (P) and multi-scale positional property pooling (MSP). The audit profile comprises paired stability, nearest-clean retrieval, grouped local-change readout and blockwise contribution profiles; feature dimension is reported separately as a resource axis. Lower-range behaviour is evaluated densely from 50 to 75 bp, whereas 100, 125 and 150 bp serve as broader anchors; full-position encodings and the canonical spaced-property (CSP) control provide boundary comparisons.

Alt text: Workflow diagram showing short reads separated into local-composition, nucleotide-property-coded and positional representation channels before diagnostic readouts and boundary checks.

empty bin is represented by zeros. Ambiguous bases contribute zero to the four nucleotide-property maps and one to the N-indicator map. The complete coordinate definitions are reported in Supplementary Table S6. MSP supplements the order-poor k-mer-frequency vector with coarse relative-position property summaries while remaining substantially lower-dimensional than full-position encodings. The default 2+3+4+6 binset was pre-specified as a coarse-to-fine relative-position summary whose finest scale remains above single-position resolution across the 50–150 bp study range. The seven-group ablation comprised K (136 dimensions), P (11), MSP (75), K+P (147), K+MSP (211), P+MSP (86) and K+P+MSP (222). Full-position composition or property matrices retained per-position information and were used as high-dimensional positional-readout references. CSP combines spaced-seed counts with property summaries and was used to test whether matching-oriented spaced-seed priors transfer into dense representation diagnostics.

Three historical handcrafted descriptor families were implemented as direct controls. The Type-I PseKNC control used trinucleotide composition ($k = 3$), three lagged correlation terms ($\lambda = 3$), weight $w = 0.05$ and the standardized Rise, Roll, Shift, Slide, Tilt and Twist dinucleotide-property tables, yielding 67 coordinates [35, 36]. The nucleotide chemical property plus accumulated nucleotide frequency (NCP+ANF) control interleaved three chemical-property indicators (ring structure, functional group and hydrogen-bond class) with accumulated nucleotide frequency at each position, yielding $4L$ coordinates for a read-length budget L [34]. The 64-dimensional PseEIIP control multiplied each normalized trinucleotide frequency by the sum of its three EIIP values [33, 38]. Because the original descriptors generally assume A/C/G/T input, the audit used declared deterministic extensions for ambiguous reads: invalid PseKNC and PseEIIP windows contributed zero composition mass, PseKNC correlation terms averaged only valid dinucleotide pairs, and N had a zero NCP code plus its own accumulated frequency. Trimmed NCP+ANF reads were N-padded to the declared source-length budget. Every historical matrix was row-normalized before the common drift and readout protocol. These deterministic extensions preserve fixed dimensions for ambiguous reads; alternative conventions may produce different values.

For mixed representations, each block was computed separately and internally L2-normalized before weighted assembly. For read s_i, let $\mathbf{K}_i$, $\mathbf{P}_i$ and $\mathbf{M}_i$ denote the reverse-complement canonical 4-mer composition block, the global nucleotide-property-coded block, and the multi-scale positional-property block, respectively. The internally normalized blocks were defined as

$$\hat{\mathbf{K}}_i = \frac{\mathbf{K}_i}{\|\mathbf{K}_i\|_2}, \quad \hat{\mathbf{P}}_i = \frac{\mathbf{P}_i}{\|\mathbf{P}_i\|_2}, \quad \hat{\mathbf{M}}_i = \frac{\mathbf{M}_i}{\|\mathbf{M}_i\|_2},$$

with zero-norm safeguards applied before normalization. The assembled compact representation was then

$$\mathbf{x}_i = \frac{1}{\sqrt{\alpha^2+\beta^2+\gamma^2}} \left[\alpha\hat{\mathbf{K}}_i\,;\,\beta\hat{\mathbf{P}}_i\,;\,\gamma\hat{\mathbf{M}}_i\right],$$

where α, β and γ are block weights and $[\cdot\,;\,\cdot]$ denotes concatenation. Unless otherwise stated, all three weights were set to 1. The block-weight audit compared equal weights with K-dominant $(2,1,1)$ and property-dominant $(1,2,2)$ settings; the separate MSP sensitivity audit evaluated $\gamma \in \{0, 0.25, 0.5, 1, 2\}$. Because the block norms are fixed before concatenation, the denominator is a fixed scalar for a given weight setting rather than a read-specific global norm. The reported global drift metric was

$$d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2\,,$$

which should be interpreted as standardized representation drift in the assembled feature space, not as a natural physical distance between commensurate biological units. Block normalization fixes the declared contribution of each block but does not make the underlying coordinates physically commensurate. This definition also gives the block-normalized distance identity

$$d^2_{\mathrm{mix}}(i,j) = \frac{1}{\alpha^2+\beta^2+\gamma^2}\left[\alpha^2\left\|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\right\|_2^2 + \beta^2\left\|\hat{\mathbf{P}}_i - \hat{\mathbf{P}}_j\right\|_2^2 + \gamma^2\left\|\hat{\mathbf{M}}_i - \hat{\mathbf{M}}_j\right\|_2^2\right].$$

For comparison with the composition-only drift $d_K(i,j) = \|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\|_2$, the mixed squared drift is lower than d_K^2 when

$$\beta^2\left(d_P^2 - d_K^2\right) + \gamma^2\left(d_M^2 - d_K^2\right) < 0,$$

where d_P and d_M are the corresponding blockwise drifts. Thus, lower property and MSP drift than K drift is a sufficient, but not necessary, condition for lower mixed drift. The diagnostic protocol takes matched clean/perturbed reads, constructs each block without fitting a predictive encoder, reports paired stability and nearest-clean retrieval, tests structured local change with grouped delta-readout, and decomposes the result through the seven-group ablation and prespecified conditional contrasts. The protocol produces a multi-axis representation profile. The value of block concatenation is assessed through dimensionality, weight, counterfactual and conditional-contribution audits.

We additionally evaluated one exploratory extension that couples canonical 4-mer tokens to relative position. The positional k-mer moment (PKM) block accumulates the first three polynomial moments of each canonical 4-mer over normalized read coordinates and compresses the resulting token-position array by deterministic signed hashing to 75 dimensions. The block-normalized extension, termed CK4P-MSP-PKM, appends PKM to K/P/MSP and uses the fixed assembly weight $\delta = 0.25$, for 297 features in total. Among the prespecified exploratory grid $\delta \in \{0.1, 0.25, 0.5, 0.75, 1\}$, $\delta = 0.25$ satisfied the stability–readout–cost screen and is reported as an exploratory Pareto point. The complete construction, resampling check and direction-sensitivity audit are reported in Supplementary Figure S15 and Supplementary Methods.

Data layers and perturbation design

Here, WGS denotes whole-genome-sequence-derived templates sampled from public assemblies. ART denotes the ART next-generation sequencing read simulator used for the external error-profile audit.

The experiments were organized into nine main data layers, with the scale of each layer reported in Table 2. The WGS perturbation layer contained 25,200 rows from 3,600 clean templates across six genera, 21 species and six read lengths; all species and NCBI assembly accessions are listed in Supplementary Table S7. Clean windows were sampled from the public assemblies with seed 303 and expanded into clean, 1% substitution, 3% N masking, 5-bp end trimming, combined 1% substitution plus 3% N masking, 1% short-indel and contiguous 6-bp mismatch conditions. The position-property ablation layer expanded this to 36,000 rows over the same template set and length grid. The lower-range continuity layer generated 76,440 global perturbation rows from 420 shared source templates at all 26 integer lengths from 50 to 75 bp, seven K/P/MSP representations and six non-clean perturbations. Its grouped local audit generated 18,720 matched template triplets, or 56,160 clean/local/nuisance variants, from 180 triplets per length-by-mode cell across four local modes. ART Illumina-like simulation contributed 55,961 matched clean/simulated pairs, represented as 111,922 rows, across 50, 60, 69, 75, 100, 125 and 150 bp [57]. Single-end reads were generated with the ART Illumina executable's default error profile, 0.005-fold coverage, SAM output and a base seed of 606 offset by read length; "paired" in this manuscript refers to each simulated read and its aligned clean template, not paired-end sequencing. The Critical Assessment of Metagenome Interpretation (CAMI) TOY low-complexity layer contained 16,342 eligible source-read groups after deterministic mate collapse and length filtering [14]. Six prespecified target-versus-background tasks were expanded to 129,600 task-specific rows at 69, 75 and 100 bp under clean, 3% N masking and 1% substitution conditions; source reads could recur as background across tasks, so these rows are not unique biological observations. The CAMI II marine probe used 4,000 anonymous source reads and 48,000 length/condition rows at 69, 75 and 100 bp [15]. The main local mutation layer used 250 triplets per analysis cell, four local modes and three read lengths. WGS-derived paired perturbations measured clean-versus-perturbed stability; the continuity layer isolated lower-range length effects on shared templates; ART added simulator-derived errors; CAMI resources supplied bounded external probes; controlled position and order tasks tested positional readout; and boundary probes tested spaced-seed transfer, context visibility and ARG/SNP limits. Experiment-specific seeds and complete argument sets are retained in the machine-readable run manifests.

The supplementary CK4P-MSP-PKM audit additionally used the public Dorsal binding-site and cis-regulatory-module sequences released by Clifford and Adami [58]. Exact known-site occurrences were located within their corresponding regulatory sequences, and 1,125 windows of 50, 75, 100 or 150 bp were labelled by whether the site occurred in the left, middle or right third of the window. The 32 source regulatory modules defined the cross-validation groups, so windows from one module did

Table 1. Representation families and their prespecified diagnostic roles. The rows distinguish compact K/P/MSP block combinations, historical handcrafted descriptors, high-dimensional positional-readout references and the spaced-seed mechanism comparator. Interpretations specify the evidence assigned to each family on its corresponding audit axis.

family	information channel	main role	main-text interpretation
CK4 / CK5	canonical local k-mer composition	composition backbone and comparator	compact token-frequency evidence, not database-linked read identity
P	global property summary block	perturbation-stable channel	tests a global nucleotide-property and auxiliary summary without local composition
MSP	positionalized nucleotide-property summaries	coarse positional channel	tests grouped local-change readout without K or global P
CK4+P	composition plus global P summary	global property baseline	tests whether the complete global summary adds to CK4
CK4+MSP	composition plus positionalized nucleotide-property signal	positional property ablation	tests whether MSP returns coarse layout information without global P
P+MSP	global plus positionalized property summaries	property-only combination	tests property stability and retrieval without K
CK4P-MSP	composition plus global and multi-scale property summaries	compact main representation	balances stability, grouped local-change readout and composition-linked retrieval
PseKNC / NCP+ANF / PseEIIP	historical composition–property, order or position descriptors	direct handcrafted-feature controls	define whether CK4P-MSP offers a distinct attributable stability–readout trade-off beyond conventional joint feature evaluation
full-position matrix	per-position token or property signal	high-dimensional positional-readout reference	tests fine positional information at higher dimensional cost
CSP	spaced-seed prior plus nucleotide-property-coded summary	spaced-seed transfer comparator	delimits where matching-oriented spaced seeds transfer to dense features

not cross folds. This probe measures motif-position accessibility within known regulatory sequences; binding-site discovery, affinity and regulatory function were outside the task definition.

For CAMI_TOY_low, read identifiers were joined to the available taxon mapping and normalized to a common NCBI taxon-identifier field. One deterministic representative was retained per `clean_read_id`; sequences shorter than 100 bp were excluded. The six most abundant eligible labels were selected before model fitting. They comprised five species-level taxa and one strain-level taxon; their NCBI Taxonomy identifiers, scientific names and ranks are reported in Supplementary Table S9 [59]. Each binary task used 1,200 target source groups and 1,200 background groups sampled across the remaining labels. Every source was then truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct` and `substitution_1pct` conditions. The resulting 21,600 rows per task were evaluated by fivefold `StratifiedGroupKFold`, with all length and perturbation derivatives of a source retained in one fold. These source-grouped tasks quantify coarse target-versus-background transfer; no fine-taxonomic endpoint was constructed.

We also added a lightweight CAMI II marine subset as an external short-read stability probe. We streamed a prefix of CAMI II marine short-read sample 0 and parsed 4,000 anonymous 150 bp reads from the embedded `anonymous_reads.fq.gz` member. These reads were truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct`, `substitution_1pct` and `substitution_1pct_N_3pct` conditions, yielding 48,000 length/condition rows. CAMI II provides benchmark truth resources, but the anonymous FASTQ headers did not provide read-level taxonomic labels in the streamed-read file used here, and the separate OTU-specific truth bundles were not reconstructed for this analysis. The CAMI II probe was therefore restricted to paired perturbation stability on an external metagenomic sequence source.

This layered simulation design was used because each component isolates a different failure mode. The controlled WGS grid makes perturbation type, read length and representation family directly comparable. The continuity layer isolates the lower read-length variable through shared-template truncation. It was analysed descriptively, whereas inferential contrasts used the prespecified discrete grid. ART adds a recognized Illumina-like simulator layer using the fixed CK4P-MSP representation. Because the ART profile produces non-monotonic cycle-dependent error loads, it supports within-length representation comparisons and quality-stratified ordering rather than a causal attribution of all changes across 50–150 bp to read length.

Metrics and statistical analysis

Perturbation stability was quantified by paired cosine similarity, paired L2 drift and nearest-clean retrieval. Retrieval searched at most 500 clean candidates within each matched analysis cell; its random-candidate chance level was therefore approximately $1/n_{\mathrm{pool}}$, and saturation at 1.0 limits its ability to rank strong K-containing representations. Representation-level readout accessibility was quantified by macro-F1 and accuracy from shallow logistic-regression or nearest-centroid probes. Except for the explicitly defined CAMI fixed-head audit, logistic probes used training-partition-standardized inputs, scikit-learn's default L2 penalty and $C = 1$, a fixed random seed, and a 70:30 stratified split for global readouts; local-change probes used fivefold stratified grouped cross-validation. The maximum iteration count was 5,000 for manuscript readouts and 4,000 in the dimensional-reduction audit. These readout metrics quantify the accessibility of retained signal to low-capacity models. Compactness was measured by feature dimension. Local mutation analysis used selective-sensitivity ratios and grouped delta-readout; all derivatives of one template were kept in the same fold through template identifiers. A selective-sensitivity ratio was treated as undefined when the matched nuisance drift was at or below 10^{-8}; valid-ratio counts and fractions

Table 2. Data layers, analysis scales, diagnostic questions, primary metrics and claim boundaries. Row counts include perturbation and task derivatives where stated and therefore are not counts of independent biological observations.

data layer	scale	diagnostic question	primary metrics	claim boundary
WGS perturbation pairs	25,200 rows; 3,600 clean templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	Which representations keep clean and perturbed reads close?	paired cosine, L2 drift, retrieval	controlled representation stability
Position-property ablation	36,000 rows; 3,600 templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	What do P and MSP add to the CK4 composition backbone?	paired stability, block drift, delta-readout, MI proxies	decomposable representation audit, not a natural physical metric
Shared-template length continuity	420 shared templates at every integer length from 50 to 75 bp; 76,440 global rows and 18,720 local triplets	Does the stability–readout profile persist across the lower short-read range?	paired drift, grouped delta-readout, conditional P/MSP increments	descriptive sensitivity over 50–75 bp
ART Illumina-like simulation	55,961 matched clean/simulated pairs; 111,922 rows; lengths 50,60,69,75,100,125,150 bp	Does the representation ordering persist within simulator lengths and quality strata?	drift ratio versus CK4, observed base difference rate	within-length simulator context; not a causal length trend or clinical error model
CAMI_TOY_low fixed-head probes	16,342 eligible source groups; 6 binary target tasks; 2,400 groups and 21,600 task rows per target; lengths 69,75,100 bp	Is coarse label information retained under length and perturbation shifts without refitting?	fixed-head macro-F1, retention, MCC, AUROC/AUPRC	source-grouped coarse-label transfer; not fine taxonomic classification
CAMI II marine anonymous-read probe	48,000 rows from 4,000 anonymous source reads; lengths 69,75,100 bp	Do stability trends persist in a more complex external metagenomic read source?	paired cosine, L2 drift, retrieval	paired stability without reconstructed read-level taxonomic labels
controlled position tasks	controlled synthetic tasks over the short-read length grid	What does fine positional resolution add?	macro-F1, feature dimension	high-dimensional positional-readout reference
local mutation sensitivity	250 triplets per analysis cell; 12 representations; 4 local modes; lengths 69,100,150 bp	Can local nucleotide-property change be detected beyond equal-count noise?	sensitivity ratio, delta-readout macro-F1	selective sensitivity, not functional effect prediction
boundary probes	spaced-seed, context-visibility and ARG/SNP boundary screens	Where do spaced-seed, context and ARG/SNP claims stop?	seed stability, visibility threshold, macro-F1	mechanism boundary and failure modes

were retained so that near-zero denominators could not produce arbitrarily large finite ratios. For the seven-group audit, three contrasts were prespecified: K+P+MSP versus P+MSP for the conditional K contribution, versus K+MSP for P, and versus K+P for MSP. Mean matched-cell differences were summarized by 10,000 paired bootstrap resamples. Two-sided paired Wilcoxon signed-rank tests were adjusted across the 12 metric-by-contrast rows by the Benjamini–Hochberg procedure [60, 61]. The analysis unit for these tests was the matched length-by-perturbation or length-by-local-mode cell. Bootstrap intervals therefore summarize variation across matched analysis cells rather than between-cohort uncertainty. Same-dimension PCA/SVD controls used randomized PCA or TruncatedSVD fitted on the training partition at 147- and 222-feature targets with the experiment seed; block-weight audits tested reasonable reweightings of the composition and property blocks.

The CAMI fixed-head audit used the block-normalized representation directly, without coordinate-wise variance scaling. Within each fold, a class-balanced L2-penalized logistic probe with $C = 1$ was fitted only to 100-bp clean training sources and then reused without refitting on every held-out length-by-condition derivative. Macro-F1, balanced accuracy, Matthews correlation coefficient, area under the receiver operating characteristic curve and area under the precision–recall curve were recorded; macro-F1 was the prespecified display metric. Source-stratified bootstrap intervals retained paired predictions within each binary task. Cross-target summaries treated the six target definitions as descriptive blocks. Because the tasks share one CAMI source pool and may reuse source reads as background, target-bootstrap intervals and paired signed-rank summaries characterize consistency among task definitions rather than between-community variability. Logistic $C \in \{0.1, 1, 10\}$ was examined as a sensitivity analysis; $C = 1$ remained the prespecified result, and no value was selected from test performance. The primary CAMI summary therefore refers to the six target/background definitions. A separate single-target preprocessing-sensitivity audit used 2,486 held-out source groups and is reported separately from the six-target aggregate.

The mixed-feature question was also tested through empirical mutual-information (MI) audits. For a perturbation label Y and representation-derived distance features X, an equal-frequency discretized screen was retained as a supplementary descriptive check. The primary analysis used a Ross/Kraskov–Stögbauer–Grassberger (KSG)-style nearest-neighbour estimator with $k = 5$ on low-dimensional continuous summaries and a discrete local-versus-nuisance label [62, 63]. It evaluated d_K, $d_K + d_P$, $d_K + d_M$ and $d_K + d_P + d_M$ across 12 local-mutation cells, with 100 shared label permutations and 100 stratified subsamples per cell. Signed joint-minus-K increments were summarized by a 10,000-resample paired-cell bootstrap and a paired Wilcoxon test; the aggregate permutation P value was calculated from the mean signed increment under the shared null permutations. These estimates are estimator-dependent empirical summaries rather than absolute physical information quantities. The seven-group contribution audit was paired with a relation audit based on rank, canonical-correlation analysis (CCA) and row-wise P/MSP alignment. Together, these analyses quantify conditional empirical roles while treating P and MSP as correlated representation blocks.

The mechanism-aligned local-change audit crossed spatial organization (contiguous versus dispersed substitutions) with substitution chemistry (property-directed versus uniformly random alternatives). It used the same seven K/P/MSP combinations and template-grouped splits. Two readouts were prespecified: spatial localization (contiguous versus dispersed) and substitution chemistry (property-directed versus random). A separate scaling audit recomputed P and MSP under the declared mappings, fixed per-property $[0, 1]$ maps and train-fit coordinate z-scores. Leave-one-property-family-out variants tested whether one P family alone determined the stability result. A descriptive coordinate-family control then partitioned P into eight nucleotide-property moments and three auxiliary coordinates (N fraction, scaled length and normalized entropy), without changing the declared 11-coordinate P block. Both subsets were evaluated in isolation and after addition to K or K+MSP under the same WGS and template-grouped local-change protocols. These analyses quantify endpoint-specific roles while limiting coordinate-level attribution.

We added a dimension-matched high-k compression audit that extends the comparison beyond short contiguous k-mer baselines. Canonical $k = 15$ signals were compressed to the 222-feature CK4P-MSP budget using hashing-trick counts and sparse random projection. These vector controls were evaluated on the same 69, 75, 100 and 150 bp perturbation grid using paired cosine similarity, L2 drift, nearest-clean retrieval and shallow readout. MinHash was assessed separately in its native estimated-versus-exact Jaccard geometry, which is distinct from the L2-vector and linear-readout analyses. This audit tests whether compact stability persists against high-specificity k-mer information under a comparable vector budget. The historical-descriptor audit used 30 matched WGS length-by-perturbation cells (five lengths and six perturbations). Its separate historical-descriptor local-readout audit used 12 template-grouped cells (three lengths and four local modes). Mean differences from CK4P-MSP were summarized by 10,000 paired-cell bootstrap resamples and two-sided paired Wilcoxon tests, with Benjamini–Hochberg adjustment across the eight historical-control contrasts.

Feature-extraction time was remeasured under one long-duration protocol spanning all manuscript-referenced Python implementations. The benchmark included CK4, CK4+P, CK4P-MSP, CK4P-MSP-PKM, three compressed $k = 15$ routes, PseKNC, NCP+ANF and PseEIIP. Every method received the same seeded batch of 10,000 clean 75-bp sequences, resampled with replacement from 600 source reads. Timed regions began with in-memory sequence strings and ended when the feature matrix was returned; disk input/output was excluded. Each route received an untimed 100-read warm-up. Methods then ran in randomized round-robin order until they reached both 120 cumulative timed seconds and five calls. Garbage collection preceded each call, and numerical libraries were restricted to one thread. Full-run medians and interquartile ranges were the primary comparative summaries. Second-half-to-first-half median ratios exposed load-dependent timing drift but were not used to rank methods. A separate 100,000-read pass per method assessed scaling. Wall-clock time was measured with `perf_counter` on an Intel Core Ultra 5 125H workstation with 31.6 GB RAM under Python 3.13.9, NumPy 2.3.5 and scikit-learn 1.7.2. These measurements characterize the tested implementations rather than hardware-independent complexity.

Reproducibility

The reference implementation and frozen result tables are maintained in the release repository. The primary API is `methods/ck4p_msp.py`; historical descriptor definitions are centralized in `methods/historical_descriptors.py`, and their common audit is executed by `experiments/audits/run_historical_descriptor_audit.py`. The unified timing entry point is `experiments/audits/run_unified_runtime_benchmark.py`. The contribution, high-k, KSG, factorial, property-scaling, MSP-sensitivity and redundancy audits are likewise executed from `experiments/audits/` and mapped to their result tables in the repository evidence map. Historical timing outputs are retained for provenance but do not define the manuscript runtime values. Analyses used Python 3.13.9, NumPy 2.3.5, pandas 2.3.3, SciPy 1.16.3, scikit-learn 1.7.2 and Matplotlib 3.10.6 on Windows 11 (build 26200), an Intel Core Ultra 5 125H processor (14 cores/18 threads) and 31.6 GiB RAM. Peak Python-managed allocation was measured in a separate pass with `tracemalloc`; this excludes memory retained solely by native-library allocators. Runtime and allocation results are machine-specific engineering measurements rather than hardware-independent complexity estimates.

Results

Global and positional property layers have related but distinct roles

We then evaluated all seven non-empty K/P/MSP combinations over 69, 75, 100 and 150 bp reads. Pure property summaries had the lowest numerical drift (P, 0.025; MSP, 0.027; P+MSP, 0.027), but they did not preserve the same nearest-clean behaviour as K-containing representations: retrieval was 0.295 for P and 0.942 for P+MSP, compared with 1.000 for CK4P-MSP. The composition block therefore traded some numerical stability for composition-linked retrieval. Relative to P+MSP, adding K increased retrieval by 0.058 (95% bootstrap CI 0.036–0.081; BH-adjusted $q = 7.3 \times 10^{-4}$) while increasing, rather than decreasing, drift.

Conditional effects differed by audit axis. Adding P to CK4+MSP reduced mean drift from 0.156 to 0.129, a matched-cell improvement of 0.027 (95% CI 0.023–0.031; $q = 2.4 \times 10^{-7}$), but did not materially change grouped delta-readout (0.9780 versus 0.9786; $q = 0.525$). Adding MSP to CK4+P produced a similar drift reduction of 0.027 (95% CI 0.023–0.032; $q = 2.4 \times 10^{-7}$) and increased grouped local-change macro-F1 from 0.904 to 0.979 (mean difference 0.074, 95% CI 0.035–0.117; $q = 7.3 \times 10^{-4}$). Adding K to P+MSP increased macro-F1 by only 0.008 and the interval included zero. These results identify metric-specific contributions under the tested grid: P primarily reduced global drift, MSP improved grouped local-change readout and K preserved composition-linked retrieval.

The redundancy audit also cautions against overinterpreting P and MSP as separable sources. Across the tested lengths, the first canonical correlation between P and MSP was near one and their first principal components were strongly aligned, as expected because both blocks are derived from the same nucleotide-property family. However, MSP had higher numerical rank than P, and row-wise P/MSP alignment after compression was incomplete. We therefore interpret the blocks as related layers at different scales, not as interchangeable evidence sources or as measurements of unrelated physical quantities.

The 2×2 mechanism audit made that distinction more specific. Conditional on CK4+P, adding MSP increased spatial-localization macro-F1 by 0.065 (95% CI 0.034–0.098; BH-adjusted $q = 9.77 \times 10^{-4}$) and substitution-chemistry macro-F1 by 0.092 (95% CI 0.059–0.130; $q = 9.77 \times 10^{-4}$). Conditional on CK4+MSP, adding P did not improve spatial localization (difference < 0.001; $q = 0.520$) but increased chemistry readout by 0.030 (95% CI 0.013–0.048; $q = 0.0315$). These results identify complementary, metric-specific contributions: MSP improved localization and chemistry readout, whereas P added a smaller chemistry increment; substantial P/MSP redundancy remained (Supplementary Figure S11 and Supplementary Table S4).

The property-scaling audit further bounded the geometric interpretation. CK4P-MSP drift was 0.128 under the declared maps and 0.132 under fixed per-property $[0, 1]$ maps, while train-fit coordinate z-scoring increased drift to 0.390. Leave-one-property-family-out P variants had a narrow drift range of 0.155–0.156. The stability pattern is therefore insensitive to reasonable fixed remapping and is not determined by one P coordinate family, but it is not invariant to arbitrary variance amplification. P and MSP should accordingly be described as encoding-value-weighted summaries (Supplementary Figure S11 and Supplementary Table S5).

An additional coordinate-family audit separated the eight nucleotide-property moments from the three auxiliary P coordinates (N fraction, scaled length and normalized entropy). The isolated P-property and P-auxiliary subsets had mean drifts of 0.023 and 0.018, respectively, and neither retained K-containing nearest-clean retrieval. Conditional on K+MSP, replacing the full P block with either subset produced similar mixed drift (0.128 for both); however, the P-property subset produced higher local substitution-chemistry readout than the P-auxiliary subset (macro-F1 0.710 versus 0.687), whereas spatial readout was similar (0.976 versus 0.975; Supplementary Table S10). Thus, the complete P block is best interpreted as a global summary channel: its overall stability effect is not uniquely attributable to one coordinate family, while its property coordinates contribute specifically to the chemistry-aligned readout.

The P/MSP contribution and relation audits are shown in Supplementary Figures S8 and S9; the mechanism and scaling audits are shown in Supplementary Figure S11. A common-protocol implementation-time comparison with historical descriptors is reported in Supplementary Figure S12.

Empirical perturbation separability supports block attribution

After identifying the metric-specific roles of K, P and MSP, we asked whether the property blocks also added perturbation-associated separability beyond CK4 distance. A KSG-style audit was applied to low-dimensional blockwise distance summaries for local change versus matched nuisance perturbation. Across 12 local-mutation cells, the mean estimate was 0.889 bits for $d_K + d_P + d_M$ and 0.752 bits for d_K alone. The mean signed empirical separability increment was 0.138 bits (paired-cell bootstrap 95% CI 0.073–0.207; paired Wilcoxon $P = 4.88 \times 10^{-4}$; aggregate label-permutation $P = 0.0099$). The increment attenuated at 150 bp, indicating length-dependent effect size within the tested grid. Under the tested perturbation grid and estimator, P/MSP distance summaries therefore improved local-versus-nuisance separability beyond the CK4 distance summary.

This estimator-dependent audit supports the block-attribution result: the property blocks contributed empirically detectable perturbation signal under the tested local-change regime, while the block-normalized metric remained standardized representation drift rather than a physical distance with commensurate units.

Estimator sensitivity is reported in Supplementary Figure S2 and Supplementary Table S3. A separate same-dimension counterfactual tested whether the result followed from adding coordinates with similar marginal scales. Across 20 WGS stability cells, sequence-linked CK4P-MSP had mean drift 0.135, compared with 0.160 after jointly permuting the P/MSP rows and 0.629 for marginally matched Gaussian controls. Across 12 grouped local-change cells, its logistic macro-F1 was 0.979, compared with 0.879 and 0.886 for the permuted and Gaussian controls, respectively. The controls preserved dimensional expansion without preserving the observed sequence-to-property mapping (Supplementary Figure S5).

MSP short-bin reliability persists across 50–75 bp

We next tested whether MSP becomes noise-dominated when short reads are divided into finer relative-position bins. The shared-template continuity audit did not show an abrupt failure across the 26 integer lengths from 50 to 75 bp. At 50 bp, mean grouped local delta-readout macro-F1 was 0.961 with the coarsest 2-bin summary and 0.988 with the complete 2+3+4+6 binset; the complete binset remained between 0.978 and 0.999 across the lower-range grid. Thus, finer relative bins increased rather than erased the structured local-change signal under this controlled construction.

The same audit separated the conditional roles of P and MSP. Across 50–75 bp, adding P to K+MSP reduced mean paired drift by approximately 0.029–0.036, whereas adding MSP to K+P increased grouped local-change macro-F1 by approximately 0.011–0.043, depending on length. CK4P-MSP drift decreased smoothly from 0.174 at 50 bp to 0.139 at 75 bp, while CK4 decreased from 0.294 to 0.235; grouped local-change macro-F1 was 0.988 and 0.993 for CK4P-MSP at the two endpoints. The stability–readout profile persisted across the controlled lower-range sweep, with effect sizes varying by length.

Performance varied gradually around $\gamma = 1$ in the two-length weight scan at 75 and 100 bp, and increasing γ to 2 reduced global drift further. We retained $\gamma = 1$ as the fixed balanced setting. Across the tested short-read grid, static MSP maintained local-change readout while remaining a coarse positional summary. MSP binset and weight sensitivity are shown in Supplementary Figure S6, and the full 50–75 bp continuity audit is shown in Supplementary Figure S13.

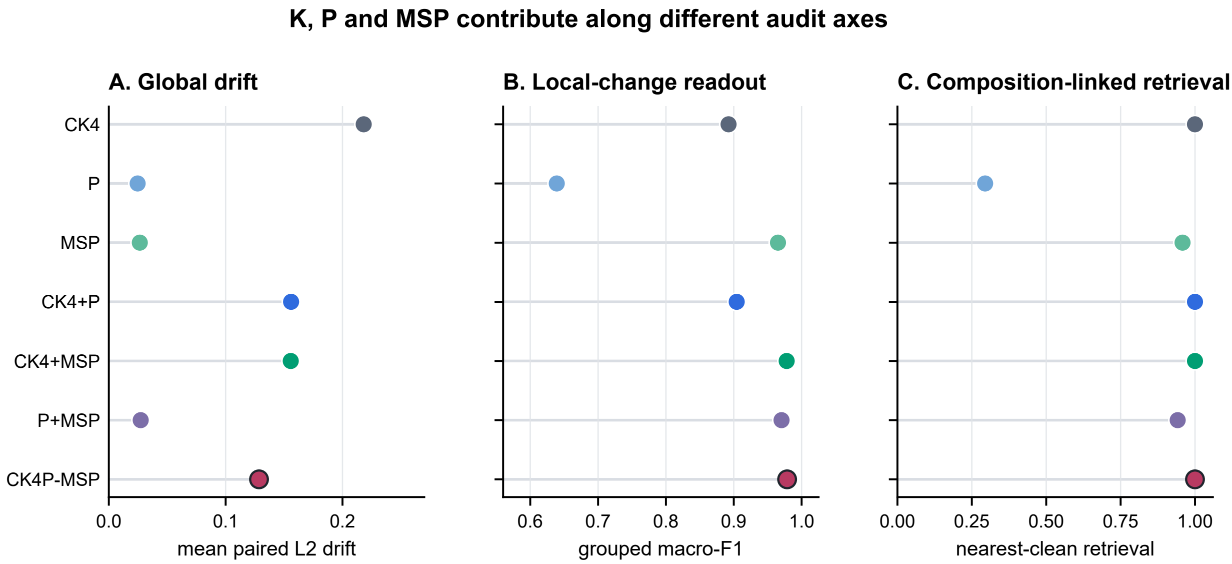

Figure 2 Seven-group K/P/MSP ablation across 24 matched length-by-perturbation cells and 12 grouped local-mutation cells. (A) Mean paired L2 drift is lowest for property-only summaries; CK4P-MSP is more stable than CK4 while retaining the K block. (B) MSP-containing representations preserve the strongest grouped local-versus-nuisance delta-readout under template-grouped fivefold cross-validation. (C) K-containing representations retain near-perfect nearest-clean retrieval from pools of at most 500 candidates, whereas P-only and P+MSP lose composition-linked retrieval. Symbols show means across cells; exact conditional contrasts, 95% paired-cell bootstrap intervals, two-sided Wilcoxon tests and Benjamini–Hochberg-adjusted q values are reported in Supplementary Table S2. These cell-level intervals and tests quantify consistency across the controlled grid; independent biological-cohort variation was not sampled.

Alt text: Three aligned dot plots compare all seven K/P/MSP block combinations by global paired L2 drift, grouped local-change macro-F1 and nearest-clean retrieval. Property-only summaries have the lowest drift, MSP-containing summaries have the strongest local-change readout, and K-containing summaries have the strongest retrieval.

Error-aware ART perturbation audit

The ART simulator audit evaluated the fixed CK4P-MSP representation and its comparators at 50, 60, 69, 75, 100, 125 and 150 bp. Within every length, CK4P-MSP retained 0.58–0.60 of the CK4 standardized drift, and CK4+P and CK4+MSP each retained approximately 0.70–0.73. The ordering was also stable within low-, middle- and high-quality strata, indicating that the property-aware reduction was not created by pooling all simulator-derived reads into one error group.

The observed same-coordinate base-difference rate was not monotonic in read length: it rose from 2.70% at 50 bp to 6.64% at 100 bp, then fell below 0.22% at 125 and 150 bp under the selected ART profile. Consequently, the ART data are used for within-length and within-stratum representation comparisons, not to infer a physical read-length response across independently generated simulator cycles. ART also remains a simulator rather than a complete model of clinical sequencing error.

Quality-stratified results and the observed ART error-load profile are shown in Supplementary Figure S3.

Compact nucleotide-property summaries reduce perturbation drift

The main compact-representation audit tested whether property-aware summaries remained closer to their clean counterparts than composition-only or compressed high-k baselines. Across the WGS-derived perturbation grid, CK4P-MSP retained paired cosine of 0.989 and mean L2 drift of 0.129 at 222 features, compared with 0.969 and 0.220 for CK4 (136 features) and 0.945 and 0.295 for CK5 (512 features). CK4+P retained paired cosine of 0.984 and drift of 0.157 at 147 features. The CK4+MSP versus CK4P-MSP contrast isolates the P-associated reduction in global drift.

Dimension-matched high-k vector controls further addressed whether a conventional high-specificity k-mer representation could show the same behaviour under the CK4P-MSP feature budget. At approximately 222 features, CK4P-MSP retained paired cosine 0.989 and L2 drift 0.129, compared with 0.821 and 0.524 for hashed $k = 15$ counts and 0.868 and 0.447 for sparse random projection of $k = 15$ counts. CK4P-MSP was more stable than the tested compressed high-k vectors, while shallow-readout performance was similar across these representations. MinHash was assessed separately through its native estimated-versus-exact Jaccard agreement (mean estimated Jaccard 0.711, exact Jaccard 0.711, mean absolute error 0.021); it was not used as an L2-vector baseline. Paired Wilcoxon tests over matched analysis cells supported the primary stability contrasts after multiple-testing correction.

Training-fitted reductions defined a separate boundary. CK7 PCA/SVD projections fitted to the clean partition at 147 or 222 target dimensions achieved mean drift of 0.093–0.098, lower than CK4P-MSP, while their logistic readout remained task-limited (macro-F1 0.403–0.416; Supplementary Figure S1). These fitted reductions define a lower-drift, data-dependent boundary, whereas CK4P-MSP provides training-free block attribution.

The fitted-reduction boundary and fixed block-weight audit are summarized in Supplementary Figure S1.

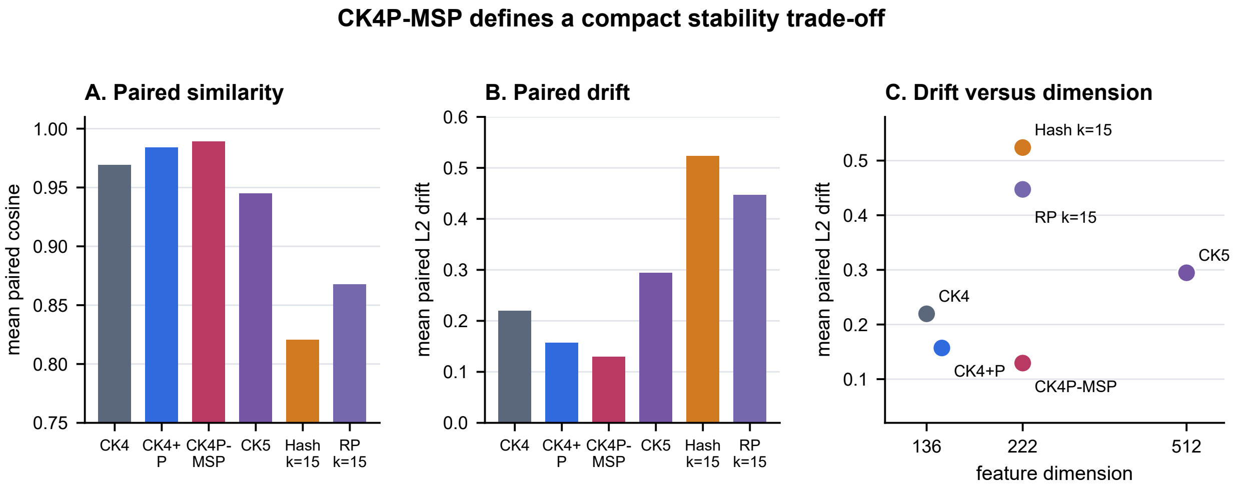

Figure 3 Dimension-matched compact stability audit over 24 length-by-perturbation cells from the controlled WGS grid. Panels compare mean paired cosine similarity, standardized L2 drift and feature dimension across compact composition/property representations and 222-dimensional hashed or sparse-random-projection $k = 15$ vectors. Bars and points show means across matched cells. CK4P-MSP is more stable than CK4, CK5 and both compressed high-k vector controls; lower drift and higher cosine indicate greater paired stability. MinHash is excluded from this L2-vector comparison and evaluated in native Jaccard geometry (Supplementary Figure S7).

Alt text: Bar plots and a scatter plot comparing compact k-mer, property-aware and dimension-matched high-k vector representations by paired similarity, standardized representation drift and feature dimension.

Compressed high-k and native MinHash results are provided in Supplementary Figure S7.

Historical descriptors define a distinct stability–readout boundary

Because composition–property fusion predates CK4P-MSP, we compared the proposed block decomposition with representative PseKNC, NCP+ANF and PseEIIP descriptors under the same paired protocol. Across 30 WGS length-by-perturbation cells, 67-dimensional PseKNC achieved the lowest mean standardized drift (0.113), compared with 0.126 for CK4P-MSP. The paired difference favoured PseKNC by 0.0133 (95% CI 0.0103–0.0163; BH-adjusted $q = 4.97 \times 10^{-8}$). CK4P-MSP therefore did not minimize either drift or dimensionality. It nevertheless had lower drift than CK4, NCP+ANF and PseEIIP by 0.0881, 0.1021 and 0.0316, respectively; all three paired intervals excluded zero after multiplicity adjustment.

The ordering reversed for the common historical-descriptor local-readout audit (three lengths by four local modes; 12 cells). CK4P-MSP reached mean logistic macro-F1 of 0.979, compared with 0.892 for CK4, 0.887 for PseKNC, 0.959 for NCP+ANF and 0.870 for PseEIIP. The paired CK4P-MSP increments were 0.086 (95% CI 0.042–0.134; $q = 5.58 \times 10^{-4}$), 0.092 (0.062–0.124; $q = 5.58 \times 10^{-4}$), 0.020 (0.005–0.036; $q = 0.0161$) and 0.109 (0.074–0.145; $q = 5.58 \times 10^{-4}$), respectively. By contrast, the separate global WGS descriptor-comparison readout averaged 0.400 for CK4P-MSP, 0.396 for PseKNC and 0.394 for PseEIIP. The 0.400 value is not the same endpoint as the 0.376 Table 3 readout, which averages the 132-cell fixed-split compact-method audit. The later PKM candidate screen likewise uses a separate candidate-route local-readout protocol and is not pooled with the 0.979 estimate. Together, these analyses locate the observed advantage in structured local-change accessibility rather than overall classification.

Under the unified long-duration benchmark, CK4P-MSP required a median 1.35 s per 10,000 75-bp reads (interquartile range 1.26–2.94 s). The corresponding medians were 1.29 s for CK4, 25.14 s for PseKNC, 1.05 s for NCP+ANF and 0.84 s for PseEIIP. Hashed $k = 15$, MinHash and sparse random projection required 2.87, 2.93 and 4.75 s, respectively. Each method accumulated at least 120 timed seconds; PseKNC required five calls, whereas faster routes required 63–157 calls. A separate 100,000-read pass required 13.09 s for CK4P-MSP and 9.19 s for CK4. The second-half-to-first-half median ratios were 2.34 and 2.35, respectively, showing that workstation load materially affected absolute wall-clock time. The complete runtime profile is reported in Supplementary Table S11. These implementation-level results place CK4P-MSP near CK4 and below the compressed high-k controls in full-run median time, with a broader IQR under sustained load. Together, the historical audit identifies a Pareto-like boundary: PseKNC is smaller and more invariant, whereas CK4P-MSP preserves stronger, explicitly block-attributable structured local-change readout (Supplementary Figure S12).

CK4P-MSP balances compactness, stability and readout accessibility

We next compared CK4P-MSP across stability, retrieval, local-change readout and dimension. Property-only variants were numerically stable but weakened nearest-clean retrieval, P improved global stability after conditioning on K+MSP, and MSP improved local-change readout after conditioning on K+P. The historical comparison showed that PseKNC achieved lower drift but lower structured local-change readout. CK4P-MSP therefore preserves a compact canonical local-composition backbone while exposing global nucleotide-property-coded and coarse positional audit channels whose contributions can be inspected separately.

Table 3. Compact main-method metrics. Paired cosine and L2 drift are means across 24 controlled WGS length-by-perturbation cells. Logistic macro-F1 is averaged across 132 fixed-split WGS readout cells and quantifies feature accessibility to a shallow probe. Feature count reports the fixed representation dimension.

representation	paired cosine	L2 drift	logistic macro-F1	features
CK4	0.969	0.220	0.363	136
CK4+P	0.984	0.157	0.364	147
CK4P-MSP	0.989	0.129	0.376	222
CK5	0.945	0.295	0.396	512

The complete representation retained K-linked retrieval, gained global stability from P and gained local-change readout accessibility from MSP. It occupies an intermediate, block-decomposable regime between composition-only compact vectors, smoother hybrid descriptors and high-dimensional full-position matrices.

External ART and CAMI probes separate stability from fixed-head label retention

ART and CAMI were used to test whether the representation family showed consistent behaviour outside the controlled WGS grid while keeping stability and label retention as separate questions. Across the seven ART lengths, CK4P-MSP retained 0.58–0.60 of CK4 drift, while CK4+P and CK4+MSP retained approximately 0.70–0.73. This within-length ratio is emphasized because the simulator error load varied non-monotonically across cycle profiles. CAMI II marine supplied anonymous-read stability. Across nine CAMI II length-by-perturbation cells, mean drift was 0.144 for CK4P-MSP, compared with 0.175 for CK4+P, 0.243 for CK4 and 0.325 for CK5.

CAMI_TOY_low supplied six labelled fixed-head tasks. Averaged over six targets and eight shifted length-by-condition cells per target, mean macro-F1 was 0.7934 for CK4P-MSP, 0.7935 for CK4, 0.7945 for CK4+P, 0.7933 for CK4+MSP and 0.7948 for CK5. These near-identical values indicate comparable coarse-label accessibility among the K/P/MSP variants. CK4P-MSP was descriptively 0.0102 macro-F1 above PseKNC and 0.0120 above PseEIIP, with the same direction in all six target definitions. Because the six tasks share one source pool, target-bootstrap intervals and signed-rank calculations summarize consistency among task definitions. Mean retention relative to each method's own 100-bp clean baseline was 0.971 for CK4P-MSP and 0.963 for CK4, whereas PseKNC and CK5 reached 0.974 and 0.972, respectively. CK4P-MSP improved retention over CK4, while PseKNC and CK5 achieved similar or slightly higher retention (Supplementary Tables S8 and S9).

The quality-stratified ART audit and the CAMI II marine subset have different evidentiary roles. The former tests the fixed representation within simulator-derived read-length and quality strata; the latter applies controlled paired perturbations to anonymous public marine reads. Marine composition can differ materially from pathogen-rich or low-biomass mNGS sources, including in GC distribution and sequence complexity. The marine analysis therefore tests whether the stability ordering persists in a composition-shifted public source; read-level taxonomic labels were not reconstructed for this probe.

Probe preprocessing created a separate boundary. In the single-target preprocessing-sensitivity audit (Supplementary Figure S14), CK4P-MSP retained mean shifted macro-F1 of 0.886 in the block-normalized input space, with a worst shifted value of 0.881 across 2,486 held-out source groups. Applying a coordinate-wise z-score fitted only on clean training reads reduced these values to 0.822 and 0.716. The primary six-target cross-target summary, which averages six task definitions, is reported separately as 0.7934 in Figure 4 and Supplementary Table S8. The largest N-mask-associated logit shifts occurred in low-variance hydrogen-bond and EIIP dispersion coordinates rather than in the explicit N-fraction coordinate. Across six targets, varying logistic regularization over $C \in \{0.1, 1, 10\}$ changed the relative ordering of the CK4-family probes, whereas CK4P-MSP remained above PseKNC and PseEIIP at all three values. Low geometric drift therefore did not guarantee invariance to learned downstream preprocessing (Supplementary Figure S14).

Together, ART supports within-length simulator consistency, CAMI_TOY_low supports coarse fixed-head label retention, and CAMI II supports anonymous-read stability. These probes extend the controlled-grid observations to public metagenomic sources at coarse fixed-head and paired-stability endpoints.

The complete CAMI II marine length-by-perturbation matrix is provided in Supplementary Figure S10; fixed-head preprocessing and regularization sensitivities are reported in Supplementary Figure S14.

Full-position matrices define a high-dimensional positional-readout reference

Full-position encodings were used to estimate what is gained when fine-grained layout information is retained. In controlled position and order tasks, full-position token or property matrices improved positional readout compared with compact summaries. This was expected, because these matrices preserve per-position structure that compact summaries intentionally compress.

Full-position matrices clarify the value of positional information at high dimensional cost. CK4P-MSP retains a coarser positional property summary at a substantially smaller feature budget, defining a compact point below this positional-readout boundary.

We evaluated CK4P-MSP-PKM to test whether more explicit token-position coupling could strengthen this boundary. In the exploratory candidate-route screen at $\delta = 0.25$, motif-position macro-F1 increased from 0.632 to 0.710 and local-readout macro-F1 from 0.888 to 0.928; these values come from the candidate screen and are distinct from the 0.979 common historical-descriptor audit above. Mean WGS drift increased from 0.129 to 0.138 and ART drift from 0.157 to 0.165. CAMI fixed-head retention was essentially unchanged (0.991 versus

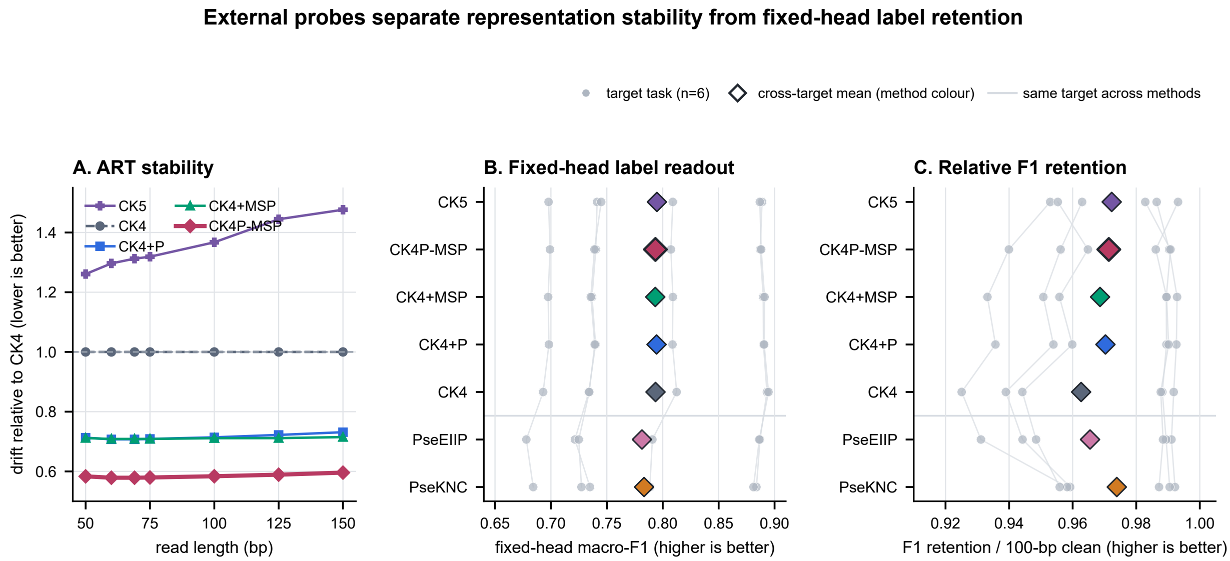

Figure 4 External probes separate simulator stability from fixed-head label retention. (A) Standardized L2 drift relative to CK4 at each ART read length; the horizontal CK4 reference is 1.0 and lower ratios indicate less within-length drift. The ART layer contains 55,961 matched clean/simulated pairs across 50, 60, 69, 75, 100, 125 and 150 bp. Lines are descriptive length-specific ratios without uncertainty intervals; the selected ART cycle profile produces a non-monotonic error load. (B) Target-level mean macro-F1 across eight shifted CAMI_TOY_low length-by-condition cells. A class-balanced logistic head was trained only on 100-bp clean sources and reused without refitting. (C) Macro-F1 retention relative to each method's own 100-bp clean baseline. Panels B and C contain six prespecified one-versus-background tasks with 1,200 target and 1,200 background source groups each. Neutral circles denote target-task estimates, method-coloured diamonds denote cross-target means and faint lines preserve target pairing across methods. The six tasks share one CAMI source pool, so cross-target summaries describe task-level consistency.

Alt text: ART drift ratios show lower relative drift for CK4P-MSP, while paired CAMI target-task panels show comparable absolute fixed-head readout among CK4-family methods and similar retention across several compact descriptors.

0.993). The extension increased dimensionality from 222 to 297 and reverse-complement-pair drift from 0.078 to 0.251. Under the unified long-duration benchmark, median extraction time increased from 1.35 s (interquartile range 1.26–2.94 s) to 2.10 s (2.03–2.39 s) per 10,000 reads, a 1.56-fold increase. A new-seed resampling check preserved the direction of the stability–readout exchange within the same benchmark resources. In the common historical-descriptor profile, CK4P-MSP-PKM exceeded PseKNC and PseEIIP on the two structured readouts but did not exceed the length-dependent NCP+ANF positional control. The variant therefore defines an optional higher-readout Pareto extension with increased drift, dimension, runtime and direction sensitivity (Supplementary Figure S15).

Local mutation analysis separates numerical stability from selective sensitivity

A stable audit representation should remain responsive to structured changes, but the relevant notion of responsiveness depends on the metric. We therefore tested whether local nucleotide-property change remained distinguishable from matched nuisance perturbation. In a distance-ratio analysis, CK4P-MSP and CK4 produced similar local-to-nuisance L2 ratios (0.752 and 0.750, respectively), whereas full-position and property-channel representations provided stronger distance amplification. In the common template-grouped audit used for the seven-group and historical-descriptor comparisons, CK4P-MSP retained high grouped delta-readout at compact dimensionality, with local-versus-nuisance macro-F1 of 0.979. The seven-group ablation assigned most of this local-readout increment to MSP, while P contributed primarily to global stability and K to nearest-clean retrieval.

These metrics resolve different properties: paired cosine and L2 drift quantify nuisance stability, distance ratios identify distance-sensitive local-change probes, and delta-readout measures whether structured local changes remain accessible to a shallow model. CK4P-MSP combines nuisance stability with decomposable local-change readout at substantially lower dimensionality than full-position matrices.

The mutation-fraction sweep is shown in Supplementary Figure S4.

Boundary analyses constrain the interpretation

Boundary probes delineated where compact descriptors were insufficient. Spaced-seed transfer remained regime dependent, context relations required visibility within the read, and ARG/SNP interpretation required database identity, annotation and allele-level evidence beyond compact property summaries.

Together, these probes separate representation-level robustness from allele-level and functional interpretation.

Table 4. Seven-group block ablation. Global drift and nearest-clean retrieval are averaged over 24 matched length-by-perturbation cells. Grouped delta-readout macro-F1 is averaged over 12 length-by-local-mode cells using template-grouped fivefold cross-validation. Lower drift and higher retrieval/readout indicate stronger performance on their respective audit axes.

representation	features	L2 drift	retrieval	grouped macro-F1	observed audit role
CK4	136	0.218	1.000	0.892	local composition and composition-linked retrieval
P	11	0.025	0.295	0.639	global property stability without local-composition retrieval
MSP	75	0.027	0.958	0.965	coarse positional property readout
CK4+P	147	0.156	1.000	0.904	composition plus global property stability
CK4+MSP	211	0.156	1.000	0.978	composition plus grouped local-change readout
P+MSP	86	0.027	0.942	0.970	stable property-only summary with incomplete retrieval
CK4P-MSP	222	0.129	1.000	0.979	balanced block-decomposable trade-off

Table 5. Boundary and mechanism summary for the spaced-seed, context-visibility and ARG/SNP probes. The table reports the tested mechanism, observed limitation and corresponding operating boundary for compact representations.

boundary	main observation	interpretation
spaced-seed transfer	The best tested 4-position dense stability pattern was contiguous 0-1-2-3 across the 69/75 bp sanity grid.	spaced seeds remain useful matching priors, but transfer is regime-specific
context visibility	motif-pair visibility thresholds depended on placement, emerging at 130, 140, 148 or 155 bp.	a representation cannot encode context absent from the read
ARG/SNP readout	stable compact features did not by themselves establish allele, resistance-SNP or functional equivalence.	identity/database evidence remains necessary for final biological calls

Discussion

This study treats short-read sequence representation as an inspectable object. Database-linked exact matching remains distinct from the compact representation itself, whereas nucleotide-property-coded and coarse positional summaries change which perturbation responses can be resolved. This positioning follows a broader alignment-free lesson that sequence length and score construction affect how similarity outputs should be interpreted [17]. CK4P-MSP occupies a compact trade-off in this setting: its reverse-complement canonical 4-mer composition block is augmented by global and multi-scale nucleotide-property channels that reduce drift relative to CK4/CK5 while retaining grouped local-change readout. Comparisons with historical descriptors place CK4P-MSP within a defined stability–readout regime.

Learning-based bimodal models may achieve stronger performance on specific biological prediction tasks when sufficient pretraining and task supervision are available, but their latent features serve a different purpose from the fixed, block-attributable coordinates evaluated here.

The results support a division of labour between representation families. Canonical k-mer counts provide a compact local-composition backbone, while exact matching and alignment retain their distinct database-linked role. Within the K/P/MSP ablation, the complete global P summary provided the strongest global stability channel. Multi-scale positional property pooling used the same property family to add coarse layout summaries to an otherwise order-poor k-mer-frequency vector and accounted for most of the grouped local-change readout gain. Full-position matrices preserve per-position structure more explicitly and define a higher-dimensional positional-readout boundary.

The exploratory CK4P-MSP-PKM extension further separates coarse layout from token-position coupling. Explicit positional k-mer moments improved motif-position and local-change readouts at the cost of higher drift, dimension, runtime and input-direction sensitivity. CK4P-MSP-PKM is therefore reported as an optional higher-readout Pareto extension.

The additional analyses sharpen this interpretation. Same-dimension PCA/SVD controls show that numerical stability is not unique to CK4P-MSP: fitted CK7 projections achieved still lower drift, establishing a data-dependent reduction boundary. The historical audit reached the same conclusion from a different direction. PseKNC was both lower-dimensional and more stable than CK4P-MSP, but its grouped local-change readout was lower. NCP+ANF approached the CK4P-MSP local readout at a larger, length-dependent dimension and higher drift. CK4P-MSP therefore occupies an attributable stability–readout regime rather than the minimum-drift corner. Dimension-matched high-k vector controls show that this fixed representation remained more stable than the tested hashed and random-projection $k = 15$ controls at a comparable feature budget, although shallow readout differences were modest. MinHash was assessed separately in its native Jaccard geometry. Block-weight and fixed-map scaling audits show that the reported trade-off is not tied to one selected weight or reasonable fixed property remapping; the deterioration under train-fit coordinate z-scoring also shows that the geometry is not scale-invariant. Paired Wilcoxon tests show that the principal contrasts are consistent across the tested matched cells. The KSG-style audit indicates an estimator-dependent empirical separability increment beyond CK4 distance in the tested local-mutation regime.

The seven-group contribution audit makes this division of labour more precise. Property-only summaries were numerically stable but did not preserve the same nearest-clean retrieval as K-containing representations. Conditional on K+MSP, P reduced global drift without a detectable increment in the original local-versus-nuisance readout. Conditional on K+P, MSP reduced global drift and produced the largest grouped local-change

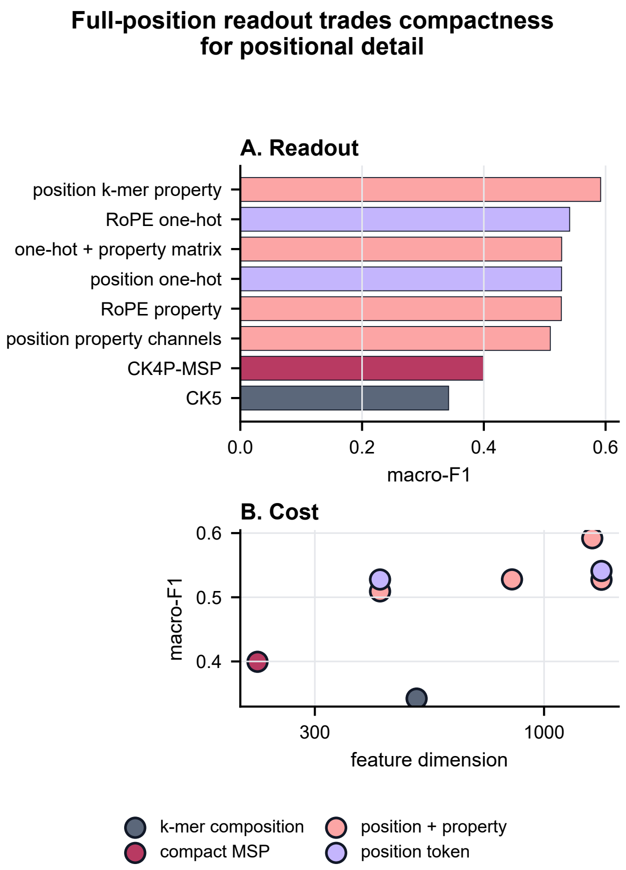

Figure 5 Full-position diagnostic reference for positional information. (A) Macro-F1 on controlled positional readout tasks compares compact summaries with representations that retain per-position token or property channels. (B) Readout is plotted against feature dimension to show the cost of preserving near-explicit positional information. CK4P-MSP occupies the compact, coarse-summary region below the full-position readout boundary.

Alt text: Bar and scatter plots showing that full-position matrices add positional readout value at substantially higher feature dimension.

readout increment. Conditional on P+MSP, K restored nearest-clean retrieval while increasing numerical drift. The factorial audit refined this assignment: MSP improved spatial-localization and chemistry readouts, whereas P added a smaller chemistry increment without a spatial increment. A coordinate-family audit showed that the full P stability effect is not unique to the eight property moments, but that those moments contributed more to chemistry-aligned readout than the N/length/entropy subset when K+MSP was held fixed; spatial readout was nearly unchanged. The redundancy audit further showed that P and MSP share a strong global property component but are not interchangeable: MSP has higher numerical rank and incomplete row-wise alignment with P. The blocks therefore remained correlated and their benefits were endpoint dependent (Supplementary Table S10).

The perturbation analyses also clarify the scope of the evidence. Uniform substitutions and local mutation sweeps provide controlled tests of representation behaviour. Across the shared-template 50–75 bp sweep, the CK4P-MSP stability–readout profile changed smoothly without a single-length discontinuity. The 100, 125 and 150 bp conditions provide broader context anchors. ART adds Illumina-like simulator context: CK4P-MSP retained approximately 58–60% of CK4 drift within each tested length and preserved the same ordering across quality strata. The selected ART profile generated non-monotonic error loads across cycle lengths, so the comparison supports within-length consistency. CAMI_TOY_low adds six source-grouped one-versus-background tasks drawn from a shared source pool. CK4P-MSP preserved shifted macro-F1 comparable to CK4/CK5 and was descriptively above PseKNC/PseEIIP in each target definition, although it did not improve absolute readout over every internal block combination or maximize relative retention. The strongest local-readout result comes from the controlled template-grouped perturbation task and therefore establishes feature accessibility under that task definition, not a general biological classification advantage. The CAMI II marine subset provides an anonymous-read stability probe without reconstructed read-level taxonomic labels. Together, these public-source probes support external stability and coarse label retention within their defined endpoints.

The boundary analyses identify where compact descriptors become insufficient. CSP and spaced-seed controls show that transfer of matching-oriented priors into dense read-level representations is regime dependent. Context and ARG/SNP probes further separate compact stability from allele-level or functional interpretation, which require database identity, annotation, sufficient context and downstream biological evidence.

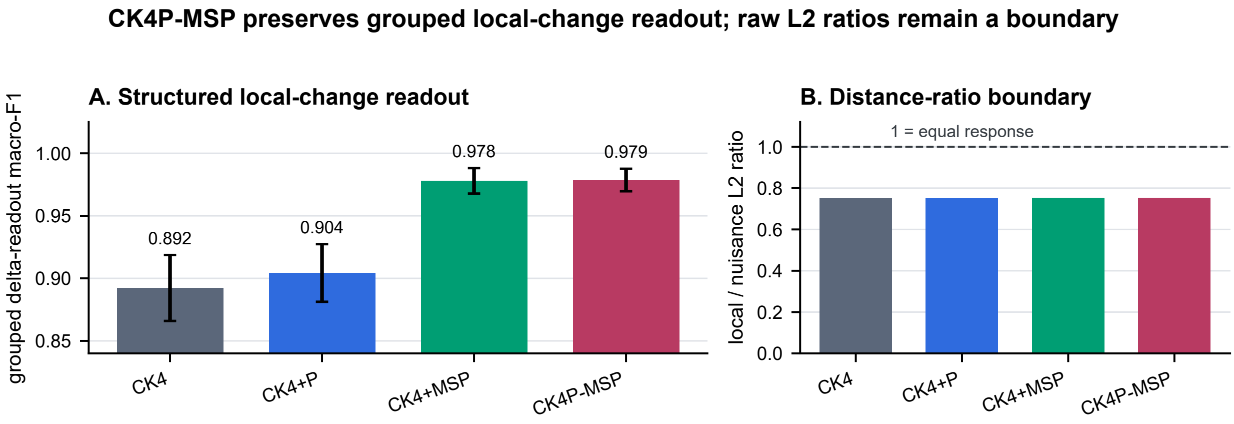

Figure 6 Common template-grouped local-change readout and distance-ratio boundary across 12 length-by-local-mode cells (250 template triplets per cell). (A) Mean local-versus-nuisance macro-F1 under template-grouped fivefold cross-validation shows that MSP accounts for most of the compact local-change readout increment. (B) CK4P-MSP and CK4 have similar mean local-to-nuisance L2 ratios, whereas full-position and property-channel representations provide stronger distance amplification. Error bars are 95% matched-cell bootstrap intervals; they quantify variation across the controlled grid, for which independent biological-cohort variation was not sampled. Higher values indicate greater readout or distance selectivity, respectively.

Alt text: Bar plots showing MSP-driven grouped local-change readout and the separate compact-representation distance-ratio boundary.

Limitations

The first limitation is task scope. The study evaluates representation diagnostics, whereas Kraken 2, Centrifuge, CLARK, Kaiju and related systems provide the relevant reference frame for database-linked taxonomic assignment. Direct integration with production classifiers, including downstream false-hit reduction or classifier-output auditing, remains to be evaluated.

The second limitation is the perturbation model and external benchmark scale. Controlled substitutions, ambiguous-base perturbations and local mutation sweeps isolate representation behaviour but do not reproduce every physical process in sequencing data. The grouped local-change readout is intentionally aligned to spatial organization and substitution chemistry, so its high score demonstrates accessibility of the tested structured perturbation signal rather than general biological utility. ART adds simulator-derived context, although its cycle-specific error load prevents treating the seven ART lengths as a clean causal length experiment. CAMI_TOY_low supports six coarse fixed-head label tasks drawn from one source pool, whereas CAMI II marine supports paired stability because read-level taxonomic labels were not reconstructed for the anonymous reads. The six CAMI target tasks are analysis blocks rather than independent communities, and source reads can recur as background across tasks. The resulting conclusions are therefore bounded to controlled short-read, simulator-contextualized and external stability/readout settings; fine taxonomic assignment, production-scale community profiling and unobserved clinical error distributions remain outside the evaluated domain.

The third limitation concerns read-length coverage. The dense shared-template audit covers the lower 50–75 bp range, while 100, 125 and 150 bp provide broader short-read anchors. Together, these conditions characterize selected short-read regimes rather than fit a continuous length-response function; individual lengths within the lower range are experimental conditions rather than biological or clinical thresholds.

The fourth limitation concerns MI estimation and coordinate attribution. The empirical MI and conditional-MI estimates characterize perturbation-associated separability for the tested feature channels, grid and estimator. Mixed L2 distance likewise quantifies standardized representation drift after block construction and normalization and has no direct interpretation in biophysical units. Its magnitude depends on the declared property maps and block weights. Property-aware distance summaries showed a positive empirical separability increment relative to composition-only and permutation controls, with weaker effects in the 150-bp cells. The P-property versus P-auxiliary audit was descriptive and was not designed to establish causal necessity for any individual coordinate family; it instead limits how specifically the global P effect can be assigned.

The fifth limitation concerns MSP weighting and scale. The short-bin audit did not show an immediate loss of reliability across the 50–75 bp sweep, and finer multi-scale pooling increased delta-readout rather than immediately degrading it. Even so, the current MSP channel remains a static coarse summary, and optimal binning and weighting across read lengths, perturbation types and sequencing platforms remain unresolved. The scaling audit further shows that train-fit variance amplification changes the geometry substantially; the reported method is the declared encoding-value-weighted representation rather than a scale-free construction.

The CK4P-MSP-PKM extension has a related but distinct limitation. Its fixed $\delta = 0.25$ value was retained from an exploratory grid and checked by new-seed resampling within the same benchmark resources. Odd positional moments preserve input direction, so the assembled extension is not reverse-complement invariant as a whole. These constraints limit CK4P-MSP-PKM to a sensitivity result pending independent validation.

The sixth limitation is computational scope. In the unified long-duration benchmark, the reference CK4P-MSP implementation required a median 1.35 s per 10,000 75-bp reads, compared with 1.29 s for CK4. Their interquartile ranges were 1.26–2.94 s and 0.93–2.17 s, respectively. A separate

100,000-read pass required 13.09 and 9.19 s. Both methods slowed during the long run, with second-half-to-first-half median ratios above two. These implementation- and hardware-specific results support an order-of-magnitude engineering comparison, not a fixed throughput guarantee. Production-scale deployment would require compiled or streaming implementations and pipeline-level cost-benefit testing.

The seventh limitation is descriptor, preprocessing and model scope. PseKNC, NCP+ANF and PseEIIP represent selected historical controls rather than an exhaustive search over handcrafted descriptor families or their task-specific parameterizations; their deterministic ambiguous-base extensions are one declared implementation choice. This selection is consistent with the broader literature, which includes both interpretable positional/compositional models and learned embeddings, but it does not establish a universal ranking of descriptor families. The fixed-head audit also showed that coordinate-wise train-fit z-scoring can amplify low-variance P coordinates under N masking even when the declared block-normalized geometry remains stable. Downstream scaling and regularization are therefore part of the readout definition. Large DNA language models and learned metagenomic classifiers may provide stronger predictive features when sufficient training data, pretraining and compute are available. A broader comparison would require a benchmark that reports predictive accuracy, preprocessing sensitivity and auditability together.

Future work

Future work should test whether CK4P-MSP or related compact summaries improve failure-mode detection when placed alongside database-centred pipelines. Pipeline-facing false-hit reduction or classifier-output auditing would require downstream classifier outputs and explicit operating-point analysis. A useful next step is to test whether composition-linked retrieval, nucleotide-property stability and grouped local-change readout can inform explicit representation-level warning signals.

A second direction is to expand the perturbation model. Platform-specific error-transition matrices, explicit Q-score trajectories and larger simulator grids would make the error-aware analysis more realistic. The current controlled simulations are useful because they isolate representation behaviour by read length, perturbation type and feature family; future work should test whether the same conclusions hold under richer platform-specific error processes. When ethically and legally feasible, restricted clinical read collections could then be used for external validation under controlled access, with public release limited to aggregate metadata and reproducible scripts.

A third direction is adaptive positional weighting. Length- or variance-aware MSP variants would alter the representation definition and require independent validation across read lengths and error regimes. The present results establish the operating profile of static MSP but leave positional-bin optimization unresolved.

The CK4P-MSP-PKM audit also identifies a more specific extension problem: positional token moments could be made strand-aware by using paired even/odd summaries, explicit orientation pooling or reverse-complement symmetrization. Such constructions would define new representations and require independent validation.

A fourth direction is to expand both historical and learned representation boundaries. Additional pseudo-composition, autocorrelation and task-specific hybrid descriptors can be evaluated under the same paired audit. Full-position matrices and DNA foundation-model embeddings are also natural candidates because they may recover richer context and stronger predictive signal than compact summaries. A shared benchmark could compare learned and handcrafted representations across predictive accuracy, composition retention, perturbation stability, positional readout and local-change sensitivity.

Conclusion

This study shows that compact short-read representations can be evaluated as block-attributable audit objects. In controlled settings, CK4P-MSP combined a canonical local k-mer-composition block with a global P summary and positionalized property channels to reduce perturbation drift relative to CK4/CK5 while preserving grouped local-change readout. PseKNC established a lower-drift, lower-dimensional historical boundary, whereas CK4P-MSP retained stronger structured local-change readout and explicit K/P/MSP attribution. The coordinate-family audit further assigned the property moments to chemistry-aligned readout without making them the unique source of mixed stability. Across six source-grouped external coarse-label tasks, CK4P-MSP retained fixed-head readout comparable to CK4/CK5 and was descriptively above PseKNC/PseEIIP; downstream scaling exposed a distinct N-mask failure mode. Compact nucleotide-property-coded and coarse position-aware summaries can therefore provide block-decomposable coordinates for representation-level perturbation and preprocessing audits alongside database-linked exact matching and curated reference methods.

Data Availability

All processed data tables, figure source summaries and analysis outputs generated for this study are maintained in the versioned GitHub repository ultrashort-dna-representation-diagnostics, on the `release` branch. The archived `v1.2.0` source and output snapshot is available through Zenodo at doi:10.5281/zenodo.21882250. Subsequent archived versions are linked by the Zenodo concept DOI doi:10.5281/zenodo.21792340.

Publicly reused resources include simulator- and benchmark-derived materials used for the shared-template WGS continuity audit, ART, CAMI_TOY_low, CAMI II marine and Dorsal motif-position analyses [57, 14, 15, 58]. The public release contains the 420-template continuity manifest, compressed perturbation summaries, local-triplet results, ART generation manifests, quality-stratified summaries and the public motif-position subset used in the analysis. The CAMI II marine stability probe used the public CAMI II marine short-read sample 0 reads archive and its corresponding setup archive; exact resource URLs are listed in the release repository. The release includes the streamed CAMI II marine subset used in the paired stability analysis; read-level taxonomic labels were not reconstructed from the separate truth resources.

Code Availability

The analysis scripts used to generate tables, figures and methodological audits are maintained in the same MIT-licensed repository. The repository README and release manifest map each manuscript figure, table and supplementary audit to its source script, input tables and generated outputs. The `v1.2.0` software snapshot is archived at doi:10.5281/zenodo.21882250; the complete version history is discoverable through doi:10.5281/zenodo.21792340.

Supplementary Data

Supplementary Data are available at *NAR Genomics and Bioinformatics* Online. The supplementary file includes Supplementary Figures S1–S15 and Supplementary Tables S1–S11. The underlying machine-readable source tables and figure-generation inputs are included in the public archived release.

Ethics and Data Governance

This study used only publicly available reference and benchmark data and computationally simulated sequences. No human participants, patient-level data or identifiable private information were involved; therefore, institutional ethics approval was not required.

Author Contributions

Ruixiang Mei: Conceptualization, methodology, software, formal analysis, investigation, visualization, writing – original draft, writing – review and editing, and project administration. Zhi Chen: Data curation and writing – review and editing. Rui Cao: Software. Xunbing Gong: Data curation. All authors reviewed and approved the final manuscript.

AI-Assisted Tools Disclosure

Generative-AI tools (OpenAI ChatGPT and Codex, Google Gemini, and Doubao) assisted with plotting-code review and with identifying language, formatting and internal-consistency issues. They were not used as sources of experimental data, references, analyses or scientific conclusions. The authors reviewed all code, figures, citations, analyses and final text and take full responsibility for the work.

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Conflict of Interest

The authors declare no competing interests.

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